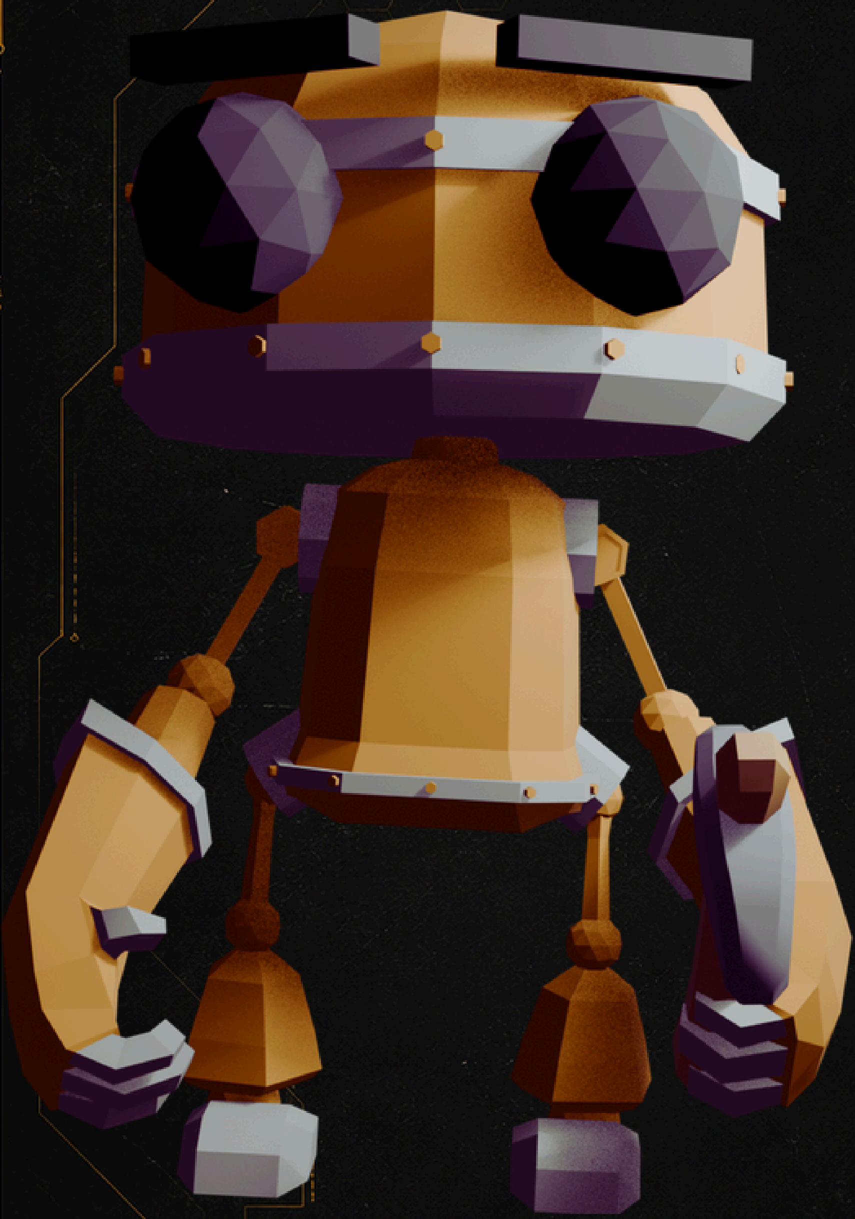
MES

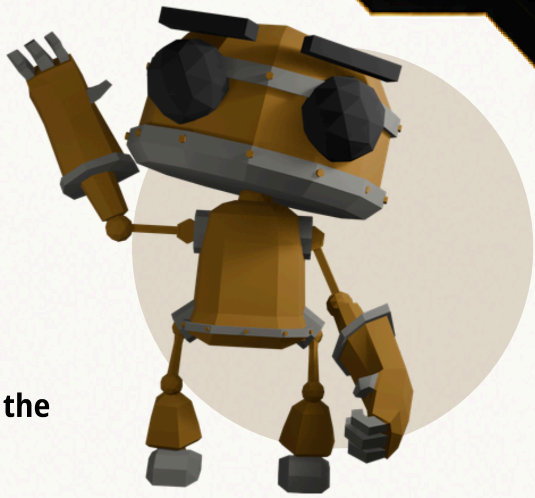


User Manual

MES

Github Repository Viewer

01 PROGRESS



Our development process followed the main goals defined at the beginning of the project:

ROADMAP

- Robot Animations
- API Integration
- Visualization
- Coherence

1 We began by focusing on the robot animations, since they were central to the visual metaphor and to the overall experience of the visualization. Once the workers could move and perform construction-related actions, we developed the surrounding 3D scene, including the building, floor, sky, and camera setup.

2 Once the 3D environment was working, we integrated the github API using a consumer-producer structure, simulating a stream of data. This stream should be hourly based, where at each step, the commits done during that hour were extracted for visualization

3 Transitions and extra animations were developed to enhance user experience and fluidity. Morover, to overcome the limitation of github API, we included the option for users to use their own API key for a better experience

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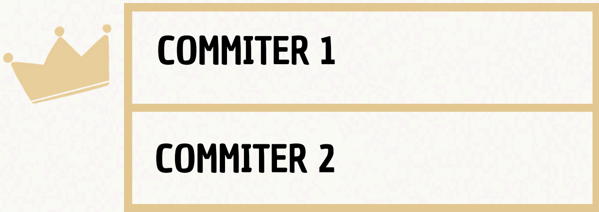
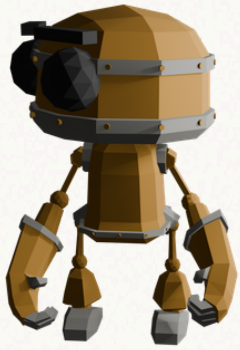
In order to display relevant information we took several approaches. We created a leaderboard for the top commiters, a general commit graph, and each worker (robot) would have its own personalized dahsboard, displaying the information about all its commits, and its personal graph.





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User experience features were added. Selected robot displays an aura. Leader board is clickable. Graph scale and view is dynamic. Each user has different colors. Background music. Firefly decoration. We realised that for big projects the building was too big, so we changed the scope, where big projects generate more than one building.

6

In the final stage, we refined the visual design of the website. We adjusted the color palette to make the scene more coherent with the interface and more comfortable to look at. We also added preference features such as a light and dark mode, allowing users to switch between two visual themes.

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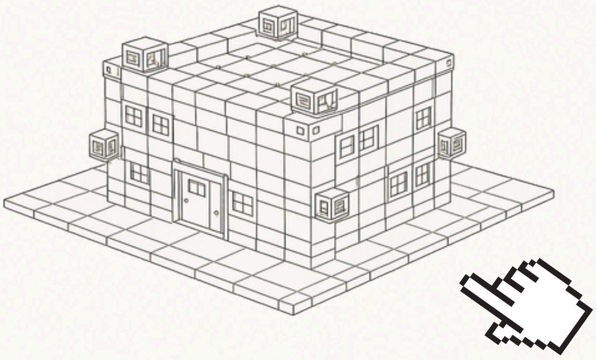
We decided to implement one of the additional goals defined earlier in the project. In particular, we added the possibility to click on individual bricks in the building wall to display the commit associated with each brick. To support this feature, we also had to revise the design of the building.

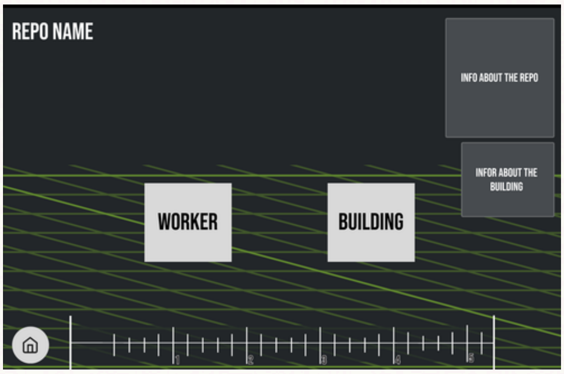


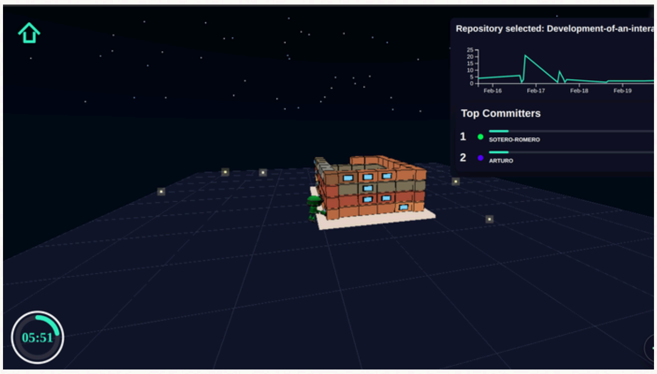
02 DESIGN EVOLUTION

SKETCHES

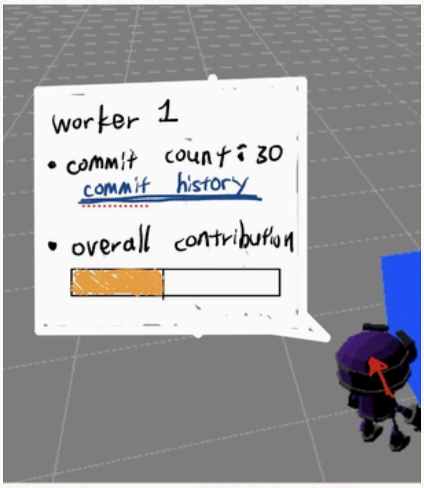
FINAL PRODUCT

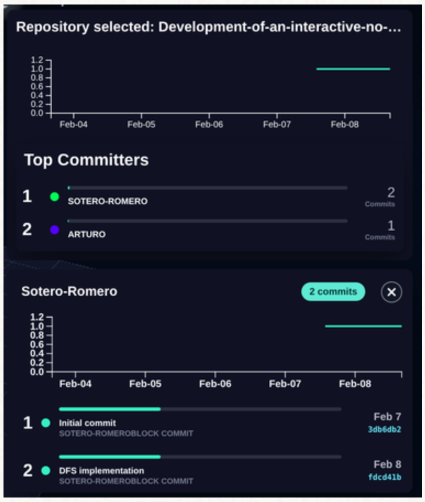
The welcome page design was pushed to the left. This allowed us to have space to add our main robot to actually welcome the user, using a wave hand animation. Color palettes also changed a bit. Yet, the overall design is mainly consistent with the inital sketch





Regarding the main page, it is also mostly consistent with the inital sketch. The “rule” based clock was changed for a normal one, the house button moved to the top, because of the clock change. Color palette changed, and the worker and robot are now realist. The information about the repository and the user was added. Notice the graph and the leaderboard





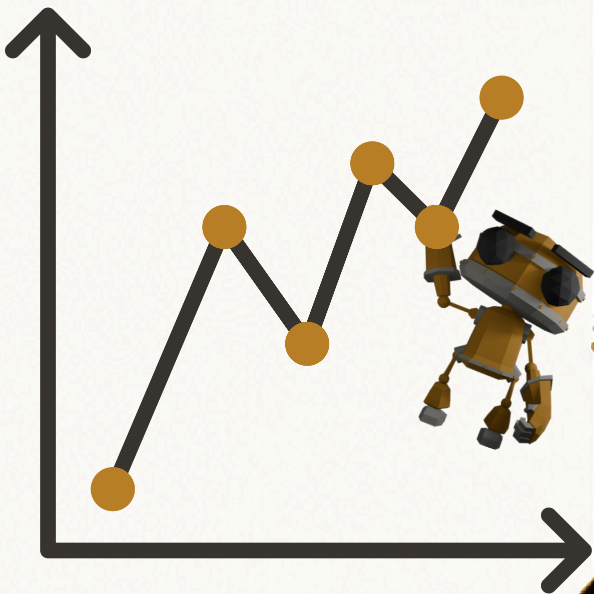
User information changed from next to the user to the left side of the screen. Overall contribution was displayed on the leaderboard. A graph displaying the user activity was added alonside the list of commits, with it respective comment, date and hash. Changes to the design were made but the overall idea remained.

03 CHALLENGES

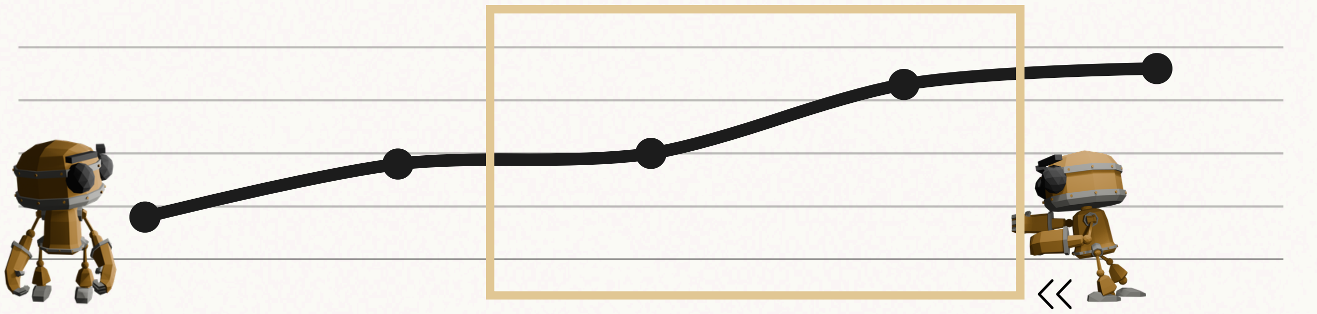
.1 COMMITS GRAPH

When implementing the graph used to display commits over time, we could not use a traditional static approach because the data arrived as a continuous stream. Since future points were unknown, the system had to remain flexible.

We considered several designs, including dynamically rescaling the graph whenever new points were added so the entire dataset stayed visible. However, this caused constant scale changes, where a single outlier could drastically alter the graph dimensions and make differences between the remaining points difficult to appreciate.



The final solution we developed was a moving-window graph with automatic resizing. This design allows users to scroll through the data while dynamically adjusting the graph size so that all points remain visible and readable. The main challenge during implementation was achieving smooth and fluid navigation throughout the graph, ensuring that points always remained visually connected while maintaining responsive and seamless scrolling behavior.

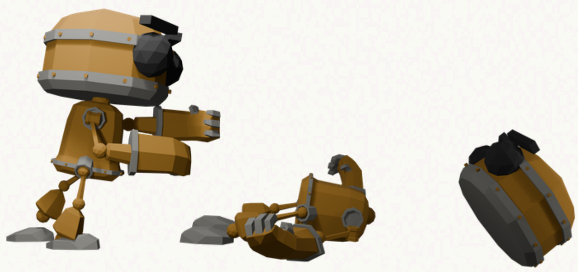


.2 SCREEN TRANSITION

Another challenge we faced, was designing a smooth transition between the welcome screen and the main visualization. Instead of loading a separate page after the user entered a repository, we decided to keep both states within the same 3D environment. This made the experience feel more continuous, but it also required careful handling of the camera, interface elements, and scene assets.

Our solution was to use the welcome screen as a close-up view of one corner of the main construction field. In this state, the camera is positioned near the scene and a welcome robot animation is shown while the repository input form is displayed. Once the user submits a repository and its data has been retrieved, the welcome-specific elements are removed or hidden, and a camera animation moves from the close-up view to a wider view of the construction site.

At the end of this transition, the main interface becomes visible and the repository visualization begins. Commits are then processed over time, robot workers are introduced into the scene, and the building progressively reflects the activity of the repository. This design decision allowed us to avoid a hard page change and instead create a more immersive transition from the introduction of the project to the data-driven visualization itself.

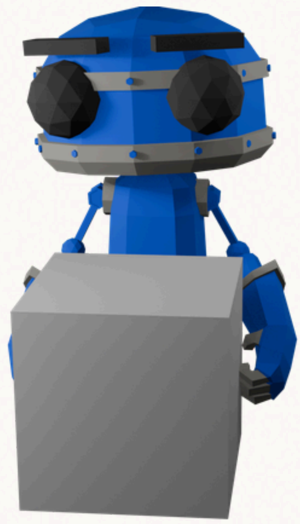


.3 COMMITS BLOCK

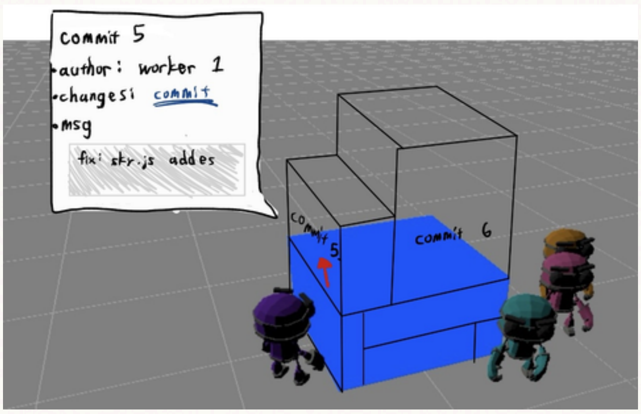
The transition from a simple block animation to an interactive building, where users can click individual bricks to inspect the commit associated with each one, was not straightforward.

Each commit had to be attached to the correct Three.js mesh through metadata so that when the user clicked a brick, the raycaster could identify the selected object and retrieve the corresponding commit.

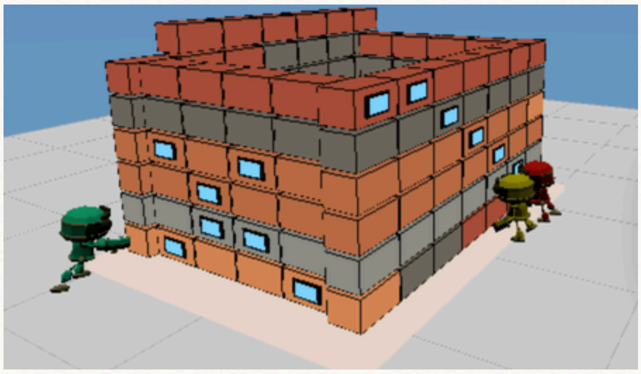
We also had to separate clickable commit bricks from non-data objects such as the foundation, roof, and ground, because clicking those parts should not open or replace the commit panel.



INITIAL IDEA



FINAL DESIGN

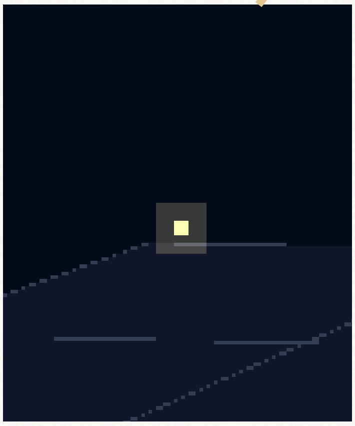


We initially considered making larger commits appear as larger blocks. However, this created several visual and technical problems: blocks could overlap, collision spacing became harder to control, and the building became less readable.

For this reason, we decided to keep all blocks the same size and use color instead. Commits from the same date share the same color, which keeps the building visually organized while still showing temporal patterns in repository activity.

.3 FIREFLY

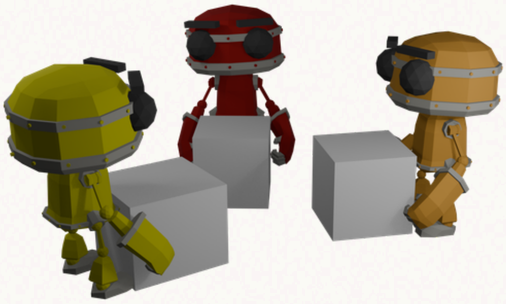
Each time a new robot enters the scene, one new firefly is added. This creates a direct link between contributor activity and the atmosphere of the visualization.



A key challenge was defining the fireflies' movement. Instead of keeping them static, we gave each firefly its own small trajectory using sinusoidal motion.Each firefly has slightly different parameters, such as speed, phase, radius, and height, so they do not all move in the same pattern.

This makes the movement feel more organic, as if the fireflies are gently floating around the scene. We also had to make sure their paths stayed within the visible area of the construction field and did not distract too much from the main visualization.

04 ASSESMENT



.1 Sotero Pedro Romero Morón

Responsible for code structure, API integration, clock, general graph, leader-board, design and merging.

.2 Mikhail Perevoznyk

Responsible for the home screen, transitions (home button) and style of the website, Helped out for the “coherence” step in the project.

..3 Sheng Wen Chen

responsible for the commit block, building animation, contributor panel (when clicking the robot), robot aura, background music, night sky, afternoon sky, setting button, and the firefly